Applications of CHEM 2263

Analytical Chemistry Cumulative Exam
October 26, 2018

There has been an increased concern about the thickness of snail shells comprised of $CaCO_3(S)$ as levels of CO_2 in the atmosphere increase. Recently a report showed a more than 50% decrease in the shell thickness for one class of snails over the past 10 years. One hypothesis for the reduction in the shell thickness would be the effect of increased water acidity due to increases in CO_2 levels. Your job in this cumulative exam is to use CHEM 2263, Analytical Chemistry I, principles to determine if such a hypothesis would be correct. Explain how you would do this with the following available information. Appropriate equations and their calculations need to be provided. The following information and constants may be applicable. For all calculations, assume that the ocean where the snail resides contains pure water.

- Consider the following aqueous equilibrium for Henry's law. In Henry's law the gas is expressed in units of atm and the aq phase is expressed in units of M.

  $CO_2(g) \leftrightarrows CO_2\ (aq)$ {all dissolved carbon dioxide in water} $K_H = 10^{-1.5}$

- For carbonic acid (H_2CO_3), K_{a1} is really defined as:

  $CO_2\ (aq)$ {all dissolved carbon dioxide} $\leftrightarrows HCO_3^- + H^+$ $K_{a1} = 4.46 \times 10^{-7}$

- HCO_3^- $K_{a2} = 4.69 \times 10^{-11}$

- $CaCO_3(s)$ $K_{sp} = 4.5 \times 10^{-9}$

- The current concentration of CO_2 is 406 ppm. 1 ppm = 10^{-6} atm.
- The concentration of CO_2 ten years ago was 380 ppm.